PianoCC: A Multimodal Dataset for Piano Coaching from Seat Pressure

Toshihiro Hirano[1,2] Ryo Nakabayashi[2] Momoko Shioki[2] Sachiko Shiotani[2] Chen-Chieh Liao[1]
Yichen Peng[1,2] Shio Miyafuji[1] Hideki Koike[1] Erwin Wu[1] Shinichi Furuya[2]
[1]Institute of Science Tokyo [2]Sony Computer Science Laboratories
{hirano.t.f29e, TODO, TODO, TODO, TODO}@m.isct.ac.jp
{TODO, TODO, TODO, TODO}@sony.com, koike@c.titech.ac.jp

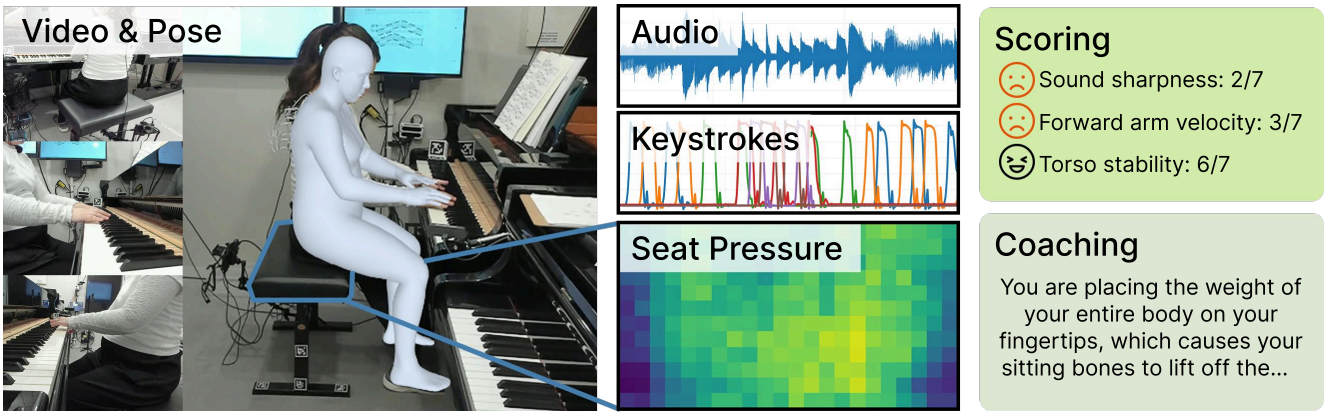


Figure 1. PianoCC synchronizes six-view video, 2D/3D pose, audio, MIDI keystrokes, and seat-pressure distributions, and connects the recorded performances to criterion-specific expert ratings and free-form expert coaching feedback.

Abstract

We present PianoCC, a multimodal dataset that connects seat pressure and body pose with piano-performance skill and expert coaching. PianoCC contains synchronized seat pressure, six-view video, 2D/3D pose, audio, and MIDI from 105 pianists performing 40 fundamental tasks, together with skill ratings and feedback text from expert instructors. We further introduce PianoCC-LM, a question-answering language model that encodes seat-pressure sequences and answers questions about the performed task, pressure and posture, skill, and possible improvements. It first aligns pressure with automatically generated descriptions and then adapts to the four QA tasks. Experiments show that pose-based system better recognizes fine-grained keyboard actions, whereas pressure-based provides complementary cues about support, weight transfer, and skill. Pressure-based outperforms pose-based in skill assessment (0.388 vs. 0.375), and combining them achieves the highest coaching AutoBCE (0.518).

1. Introduction

Piano performance depends not only on which keys are pressed, but also on how a pianist supports the body, shifts the center of mass, and transfers weight to the arms. Trunk motion affects keystroke speed, accuracy, and expression [9, 39, 40, 42], and instructors routinely address sitting posture and weight distribution. Recently, computational analyses have focused on audio, MIDI, and hand pose, while biomechanical studies have emphasized kinematics, kinetics, and electromyography [18]. However, the load transmitted through the piano stool remains largely absent from computational models of piano skill.

Existing piano datasets increasingly capture full-body motion and physiological signals [2, 8, 25, 43, 46], but none records seat pressure or pairs each performance with both criterion-specific ratings and free-form coaching. Because upper-limb actions are coordinated with trunk stabilization and weight transfer, seat pressure may complement visual kinematics by revealing how a pianist supports and redistributes body weight at the bench. In previous research,

pressure distributions have supported pose recovery in lying and seated settings [7, 32, 38], and plantar pressure has recently been connected to language models for sports coaching [14, 45]. However, it remains unknown whether seat pressure contains actionable information about seated, upper-body-dominant piano performance.

In this work, we introduce **PianoCC**, a multimodal dataset centered on seat pressure. Figure 1 summarizes its synchronized six-view RGB video, 2D/3D pose, audio, MIDI, and seat-pressure measurements, collected over 7,009 sessions from 105 pianists performing 40 fundamental conditions in 10 categories. Expert instructors additionally provide seven-criterion skill ratings and natural-language coaching for a shared task. As summarized in Table 1, PianoCC is the first piano dataset to combine seat pressure with expert ratings and coaching, and will be released to the public, including synchronized MIDI keystrokes, seat-pressure, audio, and 2D/3D pose.

We further propose **PianoCC-LM**, a language model that answers piano-related questions from seat-pressure sequences. We design an encoder that compresses the seat pressure into latent tokens, which are mapped into the language-model space by a projector. Pressure–language alignment precedes task-aware tuning for task classification, data description, skill assessment, and coaching generation. Our evaluation compares pressure with pose across the three primary tasks and ablates the alignment and multitask supervision. In our experiments, pressure obtains slightly higher skill-assessment accuracy than pose, while combining them yields the highest coaching AutoBCE.

In summary, this paper makes three contributions:

- We introduce PianoCC, a multimodal dataset of video, pose, seat pressure, MIDI, and audio with professional skill ratings and coaching text.
- We propose PianoCC-LM and a two-stage pipeline that grounds seat-pressure representations in language and answers four types of piano QA.
- We evaluate how pressure and pose differently affect task classification, skill assessment, and coaching generation, together with ablations of the proposed training design.

2. Related Work

2.1. Piano Performance and Motion Datasets

Piano datasets have developed around the relationship between sound, keystrokes, and hand motion. Motion-capture datasets provide precise 3D hand trajectories [43, 46], while recent work scales hand-pose supervision through pseudo-labels from online video [8, 20]. SKY-Piano synchronizes hand and full-body motion with audio, MIDI, and multi-view video [2]; other musical-performance datasets capture full-body motion with semantic labels [15, 21]. PianoKPM further records intrinsic hand-muscle activity. [25].

These datasets describe increasingly detailed performer motion, but omit the supporting surface through which the pianist stabilizes the trunk and transfers weight. Their supervision also focuses on motion, musical attributes, or coarse proficiency, rather than pairing specific judgments with instructions about how to improve. PianoCC complements visible body motion with seat pressure and connects both to ratings and coaching collected directly from experts.

2.2. Performance Assessment and Coaching Support

Musical-performance assessment is commonly formulated as predicting a score, rank, or specific values from video or audio [29, 30, 51]. Although such predictions quantify proficiency, they do not identify what should be changed, and limited interpretability remains a central problem in action-quality assessment [50]. Recent methods therefore generate natural-language feedback. ExpertAF derives expert-like commentary from video and 3D pose by transforming Ego-Exo4D activity commentary into weak supervision [1, 10]. CoachMe produces sport-specific instructions from deviations between learner and reference motions, but requires that reference at inference time [49]. Similar approaches extend language feedback to other skilled activities [3–5, 33, 44, 48], while general-purpose vision–language models remain insensitive to subtle differences in skill [11]. Piano-specific practice systems provide augmented visual feedback on muscle activity or expertise-dependent movement differences [19, 24, 27], but do not generate free-form coaching from the pianist's supporting-surface load.

Among these directions, pressure-based coaching is particularly relevant to our setting. Skiing datasets combine plantar pressure, pose, and expert comments [12, 45], and SoleCoach maps VQ-VAE pressure representations into a language model for coaching [14]. These studies demonstrate that load distributions can support instruction without an external camera, but focus on standing, whole-body sport motion. Piano performance is seated and dominated by upper-body actions, so relevant differences may propagate only weakly to the supporting surface. PianoCC therefore studies both criterion-specific skill assessment and open-ended coaching from seat pressure in this seated, upper-body-dominant domain.

2.3. Seat Pressure and Sensor–Language Models

Seat-pressure sensing has progressed from posture classification and everyday activity recognition [16, 36, 37] to reconstructing body pose and shape [7, 38]. ChairPose estimates full-body pose sequences from pressure and generalizes to unseen users and chairs [32], establishing that contact distributions encode more than static sitting categories. These methods, however, predict body configuration rather than its task-dependent quality or possible improvements.

Table 1. Comparison of piano-related datasets. Duration is reported per camera view; the six PianoCC views total 83 hours and 15 minutes.

Dataset	Subject	Size	Video	Audio	MIDI	Pose	Pressure	Skill Ratings
PianoHand2.5M [46]	21	5.8 h	1 view	✗	✓	Hand	✗	✗
FürElise [43]	15	10 h	5 view	✓	✓	Hand	✗	✗
PianoMotion10M [8]	unlabeled	116 h	1 view	✓	✓	Hand	✗	✗
PianoKPM [25]	20	12.6 h	3 view	✓	✓	Hand	✗	✗
SKY-Piano [2]	19	11.0 h	4 view	✓	✓	Hand / Body	✗	2 levels
PianoCC (Ours)	**105**	**13.9 h**	**6 views**	✓	✓	Hand / Body	✓	**7 items / Coaching**

Time-series sensors are commonly connected to language models by encoding the sequence and projecting its representation into the language space [6, 26]. SensorLLM first aligns signals with automatically generated trend descriptions and then performs task-aware tuning [22]. VQ-VAE tokenization has likewise supported motion understanding and coaching [17, 34, 41, 47]. PianoCC-LM adopts these established principles: a dedicated encoder preserves temporal sensor structure, description QA grounds it in language, and downstream QA connects the representation to piano tasks, skill, and coaching. Our focus is therefore the new modality–task relationship and its evaluation rather than a new language-model architecture.

3. PianoCC Dataset

PianoCC comprises a Piano Performance Collection of synchronized sensor and performance data and an Expert Coaching Collection of skill ratings and free-form feedback. Together, they connect body motion and support with performance outcomes and instructional language.

3.1. Piano Performance Collection

3.1.1. Hardware Setup

A 22×12 grid of 264 pressure sensors embedded inside the piano bench records the seated load distribution at a nominal rate of 500 Hz. Six RGB cameras capture the pianist at $1{,}920 \times 1{,}080$ pixels and 60 fps, while a Femto Bolt[1] provides 2D/3D body keypoints. Performance outcomes are recorded as per-key onset time, note, and velocity from HackKey or the piano's MIDI output, together with 48-kHz stereo audio. The modalities are synchronized by shared software triggers and timestamps. The embedded sensor array requires no worn device and therefore preserves the pianist's natural movement.

3.1.2. Participants

We recruited 105 pianists who had received specialized piano education and were either professional pianists or conservatory students (27 male and 78 female). Before recording, each participant adjusted the bench height and distance to their usual playing position, so the measured pressure reflected habitual body support rather than a prescribed posture. The collection was conducted with approval from the ethics review committee of our institution.

3.1.3. Tasks and Procedure

Participants performed 40 short, fundamental conditions in 10 categories rather than complete pieces. The conditions combine hand, articulation, register, key, tempo, and dynamics to isolate specific components of piano technique. After sensor initialization and a calibration session, participants practiced and performed each condition. We segmented sessions using the onset and offset of key presses and excluded takes containing an incorrect task after inspecting MIDI and video.

3.2. Expert Coaching Collection

3.2.1. Expert Annotators

Three experienced piano teachers (aged 30–46 years; one male and two female) independently provided skill ratings and coaching. All teachers have trained pianists who won prizes in professional competitions and have been actively engaged in piano education and research. This collection was conducted under the same institutional ethics approval as the performance collection.

3.2.2. Tasks and Procedure

The annotators evaluated the *Chord LR staccato sharp* performance from each participant. This task requires simultaneous, sharply articulated staccato chords with both hands. We selected it with a top-level pianist because differences in keystroke quality are likely to appear in trunk support and shifts in the center of mass, while holding the performed condition constant across participants.

Annotators independently and repeatedly watched six synchronized video views of each performance while listening to the corresponding audio. They scored the seven criteria in Table 2 on a 1–7 ordinal scale and then entered free-form observations, strengths, and possible improvements as text or speech. Spoken comments were automatically transcribed and corrected by the annotators. We average the three expert ratings for each criterion as skill supervision. Because the experts observed each performance from dif-

[1] https://www.orbbec.com/products/tof-camera/femto-bolt/

Table 2. Seven expert-rated skill criteria.

Sound sharpness	Attack agility
Torso stability	Finger stiffness
Forward arm velocity	L/R motion symmetry
No key-bed pressing	

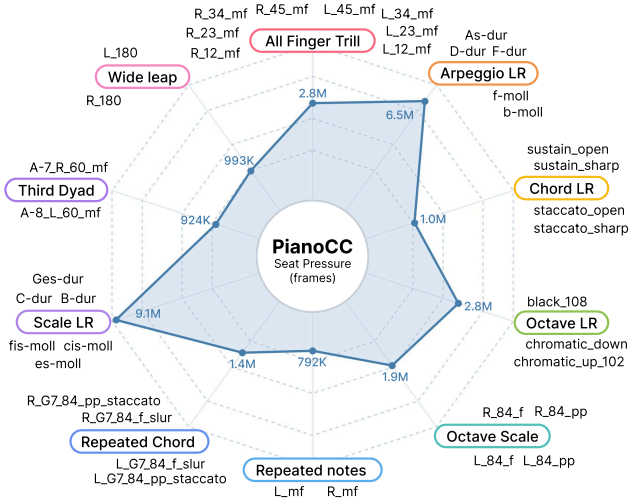


Figure 2. PianoCC task distribution. Each radial axis represents one of 10 task categories; outer labels list its fine-grained conditions, and the blue values report the number of seat-pressure frames.

ferent perspectives, we retain their coaching comments as three independent targets.

3.3. Dataset Statistics

Excluding calibration and test recordings, PianoCC contains 7,009 sessions and 13 hours and 53 minutes per camera view. The six views total 83 hours and 15 minutes of video, with 28,259,196 seat-pressure frames and 1,530,784 pose frames. Figure 2 shows all 10 categories and 40 conditions, spanning hands, registers, articulations, dynamics, and motion scales rather than concentrating on one piece or technique.

The Expert Coaching Collection covers 105 performances, 2,212 skill ratings, and 445 coaching comments. Complete three-rater scores are available for 104 performances and coaching for all 105. Each performance lasts 3.31 seconds on average, but median annotation time is one to three minutes across raters. The experts spent approximately 15 hours repeatedly inspecting each brief action from multiple views, substantially longer than its playback duration.

4. PianoCC-LM: Piano Question Answering

Having established the dataset and annotation structure, we next investigate whether seat pressure can support language-based interpretation and coaching. PianoCC-LM is a question-answering language model that takes a seat-pressure sequence and a question as input and generates a piano-related answer. It combines a seat-pressure encoder, an MLP projector, and Qwen3.5-9B [31], and is trained through pressure–language alignment followed by task-aware tuning.

4.1. Problem Formulation

Given a pressure sequence $\mathbf{S}$ and a natural-language question q, PianoCC-LM generates an answer a for one of four QA tasks (Fig. 3).

Task Classification. The model identifies one of the 40 fundamental performance conditions and answers with its task name.

Data Description. The model describes both seat-pressure behavior, including load concentration, balance, and center-of-pressure (CoP) changes, and body posture, including trunk lean, shoulder levelness, and stability.

Skill Assessment. The model answers a criterion-specific question with an integer skill rating from 1 to 7 for one of the seven criteria in Table 2.

Expert Coaching Generation. The model generates observations and actionable improvements in the style of expert piano instruction. Each expert response is treated as a separate valid target rather than merged into a single answer.

4.2. Seat Pressure Encoder

Let a seat-pressure sequence be $\mathbf{S} \in \mathbb{R}^{T\times H\times W}$, where T is its temporal length and $H \times W$ is the sensor grid. The encoder flattens the spatial grid into HW feature channels and applies temporal convolutions to compress the sequence. Its continuous, pre-quantization representation is

$$\mathbf{F}_s = \mathcal{E}_{\text{seat}}(\mathbf{S}) \in \mathbb{R}^{L(T)\times d_s}, \tag{1}$$

where $L(T)$ is the compressed temporal length and d_s is the latent dimension. ChairVAE implements $\mathcal{E}_{\text{seat}}$ as a Conv1d VQ-VAE. For variable-duration input, consecutive base windows are encoded independently and their latent sequences are concatenated in temporal order; consequently, both T and $L(T)$ vary across examples.

4.3. Seat Pressure–Language Alignment Data

Expert coaching covers only 105 performances, so we construct pressure-description QA without manual annotation. For each interval, we compute total and regional loads along the left–right and front–back axes and the CoP. Rules describe the dominant load region and discretized temporal

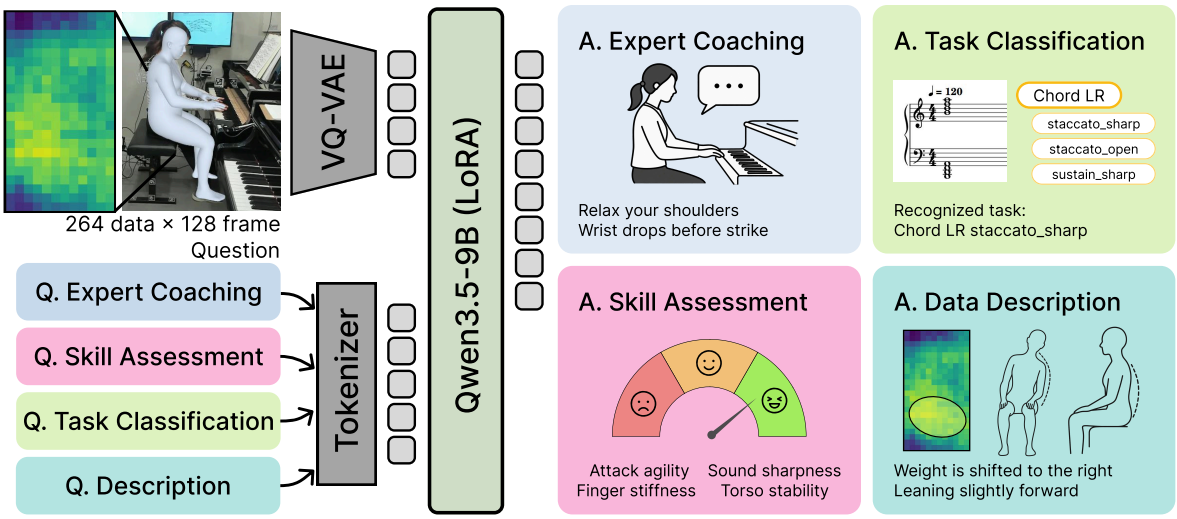


Figure 3. PianoCC-LM pipeline. A VQ-VAE encodes the seat-pressure sequence, an MLP projects its latent tokens into the language-model embedding space, and the language model performs Task Classification, Data Description, Skill Assessment, and Expert Coaching Generation in a shared QA format.

changes in these features, yielding factually grounded language about balance and weight transfer. The alignment set uses PianoCC and supplementary non-piano pressure recordings from 18 participants. These descriptions provide direct linguistic supervision for pressure properties before the model learns skill and coaching tasks.

4.4. Four-task QA Dataset Construction

After this pressure–language alignment stage, we formulate the downstream learning objectives in a unified QA format. We express all downstream supervision in a shared question–answer format that connects pressure behavior to increasingly interpretive outputs. Each example retains its source session and originating annotation identifiers. Participants are split before QA generation, preventing examples from the same pianist from crossing data splits. Stage 2 jointly samples the four tasks, allowing abundant task and description data to support the fewer expert-rated examples. See the supplementary material for details of the conversion and validation rules.

Task Classification QA. Task names in the recording manifest provide targets for the 40 fine-grained performance conditions. Each condition combines a major action category with factors such as hand, articulation, register, tempo, or dynamics. Questions ask which action was performed, and answers retain the canonical condition name in a natural-language sentence. This task tests whether the sensor representation preserves performance-dependent dynamics even when no expert annotation is available.

Data Description QA. Data-description targets cover two complementary views of body use. Pressure answers describe total and regional load, left–right or front–back imbalance, the dominant support region, and temporal CoP changes. Posture answers are generated from the pelvis, spine chest, and both shoulders in synchronized 3D pose; deterministic rules identify forward/backward and lateral lean, shoulder levelness, and trunk stability. Stage 1 uses pressure descriptions alone to establish basic pressure–language alignment. During Stage 2, pressure and posture descriptions are mixed with the other downstream tasks, teaching the model both what is directly measured at the bench and what body configuration is associated with it.

Skill Assessment QA. Skill assessment is represented as seven criterion-specific QA problems rather than a single overall proficiency label. For each performance and criterion, we average the three expert ratings, apply half-up rounding, and clip the result to an integer from 1 to 7. Separate questions for sound sharpness, attack agility, torso stability, finger stiffness, forward arm velocity, motion symmetry, and key-bed pressing preserve which aspect of technique is being judged. This format also permits criterion-level analysis of which skills are observable from pressure or pose.

Expert Coaching QA. We translate the original Japanese comments into English without adding or removing observations and retain each rater separately. For every rater–performance pair, an *overall* target preserves the complete feedback, while *individual* targets divide it into self-contained coaching elements. Clauses that would lose their meaning when separated remain together. Questions reflect the annotation genre: observation-oriented responses are prompted as performance descriptions, whereas instruction-oriented responses request both an observation and an improvement. This design preserves multiple valid emphases

instead of collapsing experts into one synthetic target.

QA Naturalization and Validation. To prevent the model from identifying tasks from template wording, we paraphrase generated QA with Qwen3.5-122B [31] while preserving required facts. Only questions are rewritten for task and skill QA, whereas both questions and answers may be naturalized for pressure and posture QA. Invalid rewrites fall back to canonical QA, and translated expert coaching is not paraphrased. See the supplementary material for details of these validation rules.

4.5. Architecture and Two-stage Training

The MLP projector maps $\mathbf{F}_s$ into the Qwen embedding space, and the projected sequence is inserted alongside the question embeddings. In Stage 1, ChairVAE and Qwen are frozen and only the projector is trained on pressure-description QA. In Stage 2, the initialized projector and LoRA parameters in Qwen are optimized jointly on the four-task QA dataset, while ChairVAE remains frozen. Both stages use causal language modeling with loss applied only to assistant-answer tokens.

4.6. Implementation Details

We resample seat pressure to 100 Hz and use a 128-frame (1.28-second) base window on the 22×12 grid. Each example contains one to three consecutive windows, producing 32–96 pressure tokens after fourfold temporal compression. Raw sensor values are converted to load, clipped by the 99.5th percentile of the training participants, and scaled to $[0, 1]$. ChairVAE uses a 512-dimensional codebook with 1,024 entries and passes its continuous pre-quantization latents to the projector. We warm up its encoder–decoder before enabling vector quantization [53, 54]. The two QA training stages run for three epochs at learning rates of 1.0×10^{-4} and 2.0×10^{-5}; Stage 2 applies rank-16 LoRA to Qwen. See the supplementary material for the loss functions and remaining training settings.

5. Experiments

We evaluate seat pressure for performance-task classification, skill assessment, and expert coaching generation. We compare it with pose to identify the information provided by each modality and ablate the proposed representation and training pipeline.

5.1. Experimental Setup

We split the 105 pianists at the participant level into 85 training, 10 validation, and 10 test participants. All methods use these splits, so no test pianist appears during training or alignment. Task classification predicts one of 40 fine-grained conditions from the recording manifest. Skill assessment treats the seven criteria as independent 1–7 classification problems and aggregates their predictions. Coaching generation uses the expert text associated with each performance. Inputs, targets, and splits are shared wherever a method supports the corresponding task.

5.2. Baselines and Comparison Models

Random Forest. We train Random Forests on summary and temporal-difference features from pressure cells and, separately, from 27 whole-body pose joints. This comparison tests conventional handcrafted representations.

Transformer. A four-layer Transformer encoder directly processes either the pressure sequence or the 27-joint pose sequence. It uses class-weighted cross-entropy for task classification and Huber loss for skill assessment.

Nearest Neighbor. This retrieval baseline finds the most similar training performance in the same modality and returns its task, skill label, or expert coaching. It tests how far direct reuse of the collected examples can proceed without generative learning.

Qwen3.5-9B. We render pressure or pose as video and provide it either to the pretrained model without adaptation or to a model fine-tuned on the same downstream QA data. These conditions test whether a generic visual encoder can replace a dedicated sequence encoder.

PianoCC-LM (Pose). We construct a cross-modal counterpart by replacing ChairVAE with a VQ-VAE for 3D pose while retaining the projector, language model, and two-stage training. Unlike a component ablation, this model asks which aspects of task, skill, and coaching are conveyed by seat pressure rather than externally observed body motion. A combined variant tests whether the modalities provide complementary information.

5.3. Evaluation Metrics

We report accuracy and macro-F1 for 40-class task classification and accuracy for seven-level skill assessment. For coaching, BLEU [28], ROUGE-L [23], and BERTScore [52] measure surface and semantic similarity.

We additionally use Automatic Body-part Coaching Evaluation (AutoBCE) [14], because global text similarity can miss whether feedback addresses the correct body region or improvement. An LLM first decomposes reference and generated coaching into body-region-specific elements. Natural-language inference then classifies each element against the other complete response as entailment, contradiction, or neutral. Precision and recall measure the consistency and coverage of the generated elements, and their harmonic mean gives AutoBCE-F1.

5.4. Main Results and Modality Analysis

Task Classification. Table 3 shows that pose is substantially more effective for fine-grained task recognition. Its Random Forest reaches 0.646 accuracy, while the combined model reaches 0.642 accuracy and the highest macro-F1 of

Table 3. Quantitative comparison on task classification, skill assessment, and coaching generation. Bold indicates the highest value in each column.

Method	Input	Task Acc.	Task F1	Skill ACC.	BLEU	R-L	BERT	AutoBCE
Random Forest	Pose	**0.646**	0.620	0.333	–	–	–	–
Transformer	Pose	0.614	0.589	0.341	–	–	–	–
Nearest Neighbor	Pose	0.382	0.329	0.262	0.070	0.192	0.886	0.162
Qwen3.5-9B	Pose	0.022	0.022	0.301	0.000	0.104	0.881	0.319
Qwen3.5-9B-FT	Pose	0.412	0.342	0.294	0.221	0.429	0.902	0.450
Random Forest	Pressure	0.404	0.356	0.318	–	–	–	–
Transformer	Pressure	0.337	0.283	0.363	–	–	–	–
Nearest Neighbor	Pressure	0.232	0.232	0.286	0.063	0.208	0.886	0.136
Qwen3.5-9B	Pressure	0.026	0.003	0.276	0.007	0.111	0.880	0.265
Qwen3.5-9B-FT	Pressure	0.114	0.080	0.285	0.173	0.352	0.912	0.346
PianoCC-LM (Ours)	Pose	0.632	0.619	0.375	**0.247**	0.519	0.931	0.466
PianoCC-LM (Ours)	Pressure	0.390	0.354	**0.388**	0.238	**0.531**	**0.935**	0.492
PianoCC-LM (Ours)	Pressure+Pose	0.642	**0.622**	0.382	0.244	0.524	0.932	**0.518**

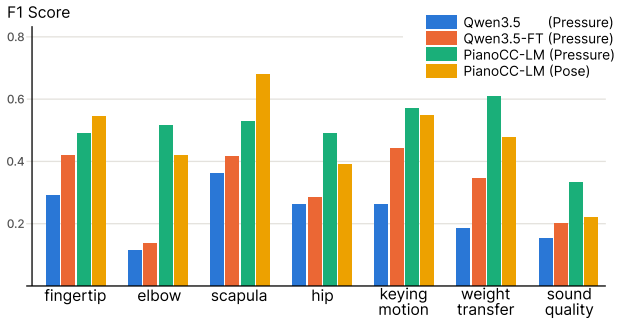


Figure 4. AutoBCE-F1 by coaching element. Pressure and pose capture different body regions and aspects of piano technique; higher values indicate closer agreement with expert feedback.

0.622. Pressure methods remain near 0.40 accuracy because related articulation or register conditions may produce similar support patterns. However, pressure-based PianoCC-LM reaches 0.752 accuracy on the 10 categories, retaining broader action dynamics without resolving exact keyboard motion.

Skill Assessment. Pressure-based PianoCC-LM obtains a skill-assessment accuracy of 0.388, compared with 0.375 for pose and 0.382 for their combination. Conventional models also show no uniform modality advantage: pose scores higher for the Random Forest, whereas pressure scores higher for the Transformer. Thus, strong task recognition does not necessarily imply strong skill assessment. Pose measures visible arm motion, while pressure adds cues about trunk stability, load distribution, and weight transfer.

Coaching Generation. Combining pressure and pose yields the highest AutoBCE of 0.518, compared with 0.492 for pressure and 0.466 for pose, directly supporting their complementary coaching cues. Fine-tuned Qwen remains below the corresponding dedicated encoder for both pressure (0.346 vs. 0.492) and pose (0.450 vs. 0.466), suggesting that the VQ-VAEs better preserve modality-specific temporal variation. Among the single-modality models, pose has slightly higher BLEU, whereas pressure has higher ROUGE-L, BERTScore, and AutoBCE. Figure 4 further shows higher pressure-based scores for weight transfer and support and higher pose-based scores for visible shoulder and fingertip cues. Among pressure-only outputs, 44.4% of coaching elements entail the expert feedback, 13.0% contradict it, and 42.6% are neutral.

5.5. Analysis of the QA Training Design

Pressure–Language Alignment. As shown in Table 4, Stage 1 provides a consistent initialization benefit. Relative to the otherwise complete model trained without Stage 1, it raises task accuracy from 0.338 to 0.390, skill accuracy from 0.375 to 0.388, and AutoBCE from 0.452 to 0.492. Improvements also appear in macro-F1 and all three text-similarity metrics. Thus, automatically generated pressure descriptions do more than add training samples: they first constrain the projector to express load patterns in a representation that the language model can use.

Data Description Supervision. The two no-Stage-1 variants separate alignment pretraining from description QA during downstream tuning. Adding Data Description to task, skill, and coaching raises task accuracy from 0.254 to 0.338, skill accuracy from 0.286 to 0.375, and AutoBCE from 0.423 to 0.452. Description supervision therefore remains useful even without pre-alignment. It provides an intermediate target between raw pressure and subjective expert judgments, encouraging the model to retain load balance, movement, and posture information that is shared by the three evaluated tasks.

Cross-task Transfer. Adding skill QA to description and

Table 4. Effects of pressure descriptions, downstream task supervision, and Stage 1 pressure–language alignment.

Variant	Desc.	Task	Skill	Coaching	T-Acc.	Task F1	S-ACC.	B@4	R-L	BERT	AutoBCE
Description + Task	✓	✓			0.262	0.233	0.306	0.004	0.153	0.883	0.078
+ Skill	✓	✓	✓		0.342	0.303	0.348	0.009	0.166	0.881	0.123
No Description		✓	✓	✓	0.254	0.212	0.286	0.210	0.503	0.929	0.423
All w/o Stage 1	✓	✓	✓	✓	0.338	0.298	0.375	0.230	0.529	0.931	0.452
All (Ours)	✓	✓	✓	✓	**0.390**	**0.354**	**0.388**	**0.238**	**0.531**	**0.935**	**0.492**

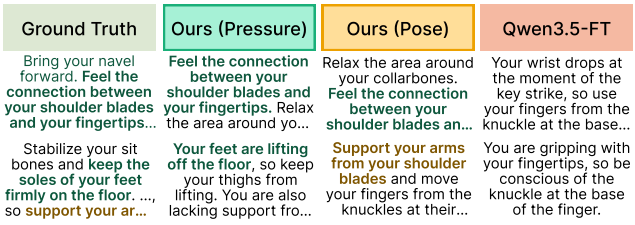


Figure 5. Expert coaching and outputs from PianoCC-LM using pressure or pose and from Qwen3.5-9B fine-tuned on pressure heatmaps. Pressure emphasizes support and lower-body load, whereas pose emphasizes visible shoulder and arm motion.

task QA improves task accuracy from 0.262 to 0.342 and skill accuracy from 0.306 to 0.348, indicating positive transfer between recognizing an action and judging how it was executed. Coaching metrics remain low when coaching supervision is absent. Adding the expert coaching task raises AutoBCE to 0.452 even before Stage 1 is restored. The complete model combines both sources of improvement. These results support the four-task formulation: automatically derived tasks build sensor understanding, but expert annotations remain necessary for specialized assessment and coaching generation.

5.6. Qualitative Results

Figure 5 illustrates the complementary modalities. Pressure-based PianoCC-LM relates changes in support to the feet, thighs, and transfer through the torso, while the pose variant directly describes shoulder and arm use. Both show partial agreement with the collected expert feedback while emphasizing different aspects of the performance. In contrast, Qwen3.5-9B-FT focuses on visible heatmap patterns and local hand mechanics less reliably grounded in the pressure signal.

6. Limitations and Future Work

Dataset. PianoCC focuses on formally trained pianists performing short fundamental tasks. This controlled setting helps associate pressure changes with specific techniques, but underrepresents the characteristic errors and compensatory movements of beginner and intermediate players. Short tasks also omit piece-level factors such as phrasing, fatigue, and transitions between techniques. Moreover, the coaching subset covers one performance condition and three experts. Future collections should therefore broaden skill levels, repertoire, playing duration, tasks, and annotators. Pressure distributions may also vary with body size, sitting position, bench geometry, and sensor placement. Evaluation across users, chairs, and sensor configurations is necessary before treating the present patterns as device-independent indicators of technique.

PianoCC-LM. Seat pressure reveals body support, center-of-mass shifts, and trunk or lower-body motion, but only weakly observes fingering, individual keystrokes, and the resulting sound. Pose instead exposes visible upper-body motion. Their fusion achieves the highest AutoBCE, supporting complementary visibility, but does not improve every skill or text-similarity metric. Given the small single-modality gaps and 10 test participants, repeated splits and confidence intervals are needed. A practical coaching system should therefore select or combine pressure, pose, audio, and MIDI according to the target aspect of technique rather than assume that adding modalities always helps. PianoCC-LM also infers desirable behavior from its training examples without an explicit reference performance. Reference-conditioned methods could directly model deviations from pedagogically validated motion or pressure [13, 49], while retrieval could ground feedback in validated examples when ideal execution depends on musical context. Finally, BLEU, ROUGE-L, BERTScore, and AutoBCE measure agreement with collected comments but do not establish factual correctness, pedagogical safety, or subsequent improvement. These outcomes require blinded expert assessment and longitudinal studies with learners.

Beyond piano, an unobtrusive pressure sensor embedded in a chair could support seated-motion description and assistance without obstructing the user or recording appearance. Compared with cameras, it may preserve more visual privacy while retaining cues about lower-body motion, weight transfer, and trunk control. Potential applications include office ergonomics, rehabilitation, and daily activity support. Such deployment nevertheless introduces domain shifts across users and furniture, and bodily pressure remains sensitive personal data. Practical systems therefore require explicit consent, limited retention, and clear restrictions on secondary use.

7. Conclusion

We introduced PianoCC, a multimodal dataset that pairs seat pressure and pose with piano performance, expert skill ratings, and coaching, and PianoCC-LM, a QA model that interprets piano performance from seat pressure. Pose better recognizes fine-grained keyboard actions, whereas pressure captures complementary support and weight-transfer information, and scores higher for skill assessment. Combining both yields the highest coaching AutoBCE, supporting seat pressure as an additional signal for piano-skill analysis and coaching.

Acknowledgments

This work was supported by JST CRONOS (Grant Number JPMJCS24N8), JST ASPIRE (Grant Number JPMJAP2404), and JST BOOST (Grant Number JPMJBS2430). This study was carried out using the TSUBAME4.0 supercomputer at the Institute of Science Tokyo.

References

[1] Kumar Ashutosh, Tushar Nagarajan, Georgios Pavlakos, Kris Kitani, and Kristen Grauman. Expertaf: Expert actionable feedback from video. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 13582–13594, 2025. 2

[2] Joonhyung Bae, Dawon Park, Taegyun Kwon, Yoon-Seok Choi, Hyeon Hur, Satoshi Obata, Shigeru Kai, Yohei Wada, Yu Takahashi, Akira Maezawa, Jaebum Park, Jonghwa Park, and Juhan Nam. SKY-Piano: A multimodal piano performance dataset. In *Proceedings of the 27th International Society for Music Information Retrieval Conference (ISMIR)*, Abu Dhabi, UAE, 2026. 1, 2, 3

[3] Edoardo Bianchi and Antonio Liotta. Skillformer: Unified multi-view video understanding for proficiency estimation, 2025. 2

[4] Edoardo Bianchi and Antonio Liotta. Parameter-efficient multi-view proficiency estimation: From discriminative classification to generative feedback, 2026.

[5] Edoardo Bianchi, Jacopo Staiano, and Antonio Liotta. Profvlm: A lightweight video-language model for multi-view proficiency estimation. *Computer Vision and Image Understanding*, 268:104749, 2026. 2

[6] Mingyue Cheng, Yiheng Chen, Qi Liu, Zhiding Liu, Yucong Luo, and Enhong Chen. Instructime: Advancing time series classification with multimodal language modeling. In *Proceedings of the Eighteenth ACM International Conference on Web Search and Data Mining*, page 792–800, New York, NY, USA, 2025. Association for Computing Machinery. 3

[7] Henry M Clever, Zackory Erickson, Ariel Kapusta, Greg Turk, Karen Liu, and Charles C Kemp. Bodies at rest: 3d human pose and shape estimation from a pressure image using synthetic data. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 6215–6224, 2020. 2

[8] Qijun Gan, Song Wang, Shengtao Wu, and Jianke Zhu. Pianomotion10m: Dataset and benchmark for hand motion generation in piano performance. In *The Thirteenth International Conference on Learning Representations*, 2025. 1, 2, 3

[9] Werner Goebl and Caroline Palmer. Temporal control and hand movement efficiency in skilled music performance. *PLOS ONE*, 8(1):1–10, 2013. 1

[10] Kristen Grauman, Andrew Westbury, Lorenzo Torresani, Kris Kitani, Jitendra Malik, Triantafyllos Afouras, Kumar Ashutosh, Vijay Baiyya, Siddhant Bansal, Bikram Boote, Eugene Byrne, Zach Chavis, Joya Chen, Feng Cheng, Fu-Jen Chu, Sean Crane, Avijit Dasgupta, Jing Dong, Maria Escobar, Cristhian Forigua, Abrham Gebreselasie, Sanjay Haresh, Jing Huang, Md Mohaiminul Islam, Suyog Jain, Rawal Khirodkar, Devansh Kukreja, Kevin J Liang, Jia-Wei Liu, Sagnik Majumder, Yongsen Mao, Miguel Martin, Effrosyni Mavroudi, Tushar Nagarajan, Francesco Ragusa, Santhosh Kumar Ramakrishnan, Luigi Seminara, Arjun Somayazulu, Yale Song, Shan Su, Zihui Xue, Edward Zhang, Jinxu Zhang, Angela Castillo, Changan Chen, Xinzhu Fu, Ryosuke Furuta, Cristina González, Prince Gupta, Jiabo Hu, Yifei Huang, Yiming Huang, Weslie Khoo, Anush Kumar, Robert Kuo, Sach Lakhavani, Miao Liu, Mi Luo, Zhengyi Luo, Brighid Meredith, Austin Miller, Oluwatumininu Oguntola, Xiaqing Pan, Penny Peng, Shraman Pramanick, Merey Ramazanova, Fiona Ryan, Wei Shan, Kiran Somasundaram, Chenan Song, Audrey Southerland, Masatoshi Tateno, Huiyu Wang, Yuchen Wang, Takuma Yagi, Mingfei Yan, Xitong Yang, Zecheng Yu, Shengxin Cindy Zha, Chen Zhao, Ziwei Zhao, Zhifan Zhu, Jeff Zhuo, Pablo Arbeláez, Gedas Bertasius, Dima Damen, Jakob Engel, Giovanni Maria Farinella, Antonino Furnari, Bernard Ghanem, Judy Hoffman, C. V. Jawahar, Richard Newcombe, Hyun Soo Park, James M. Rehg, Yoichi Sato, Manolis Savva, Jianbo Shi, Mike Zheng Shout, and Michael Wray. Ego-exo4d: Understanding skilled human activity from first- and third-person perspectives. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 19383–19400, 2024. 2

[11] Rui Henriques, Ricardo Rei, Pedro Henrique Martins, et al. Can vision language models judge action quality? an empirical evaluation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8874–8883, 2026. 2

[12] Toshihiro Hirano, Yuki Tabei, Yichen Peng, Chen-Chieh Liao, Erwin Wu, and Hideki Koike. Skitechcoach: A multimodal alpine skiing dataset with 3d body pose, sole pressure, and expert coaching. In *Proceedings of the 8th International ACM Workshop on Multimedia Content Analysis in Sports*, page 39–46, New York, NY, USA, 2025. Association for Computing Machinery. 2

[13] Toshihiro Hirano, Hitoshi Yoshihara, Chen-Chieh Liao, Erwin Wu, and Hideki Koike. Reference-based positive ski coaching using foot pressure and mllms. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops*, pages 452–462, 2026. 8

[14] Toshihiro Hirano, Hitoshi Yoshihara, Yichen Peng, Chen-

Chieh Liao, Erwin Wu, and Hideki Koike. Solecoach: Sole pressure and imu-based mllms for skill coaching. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, New York, NY, USA, 2026. Association for Computing Machinery. 2, 6

[15] Yu-Fen Huang, Nikki Moran, Simon Coleman, Jon Kelly, Shun-Hwa Wei, Po-Yin Chen, Yun-Hsin Huang, Tsung-Ping Chen, Yu-Chia Kuo, Yu-Chi Wei, Chih-Hsuan Li, Da-Yu Huang, Hsuan-Kai Kao, Ting-Wei Lin, and Li Su. Mosa: Music motion with semantic annotation dataset for cross-modal music processing. *IEEE/ACM Trans. Audio, Speech and Lang. Proc.*, 32:4157–4170, 2024. 2

[16] Nguyen Thi Minh Huong, Vo Quoc Bao, Nguyen Trung Hau, and Huynh Quang Linh. A smart chair for health monitoring in daily life. *arXiv preprint arXiv:2410.01459*, 2024. 2

[17] Biao Jiang, Xin Chen, Wen Liu, Jingyi Yu, Gang Yu, and Tao Chen. Motiongpt: human motion as a foreign language. In *Proceedings of the 37th International Conference on Neural Information Processing Systems*, Red Hook, NY, USA, 2023. Curran Associates Inc. 3

[18] Aljoša Jurinić, Marija Pranjić, Aiyun Huang, Timothy A. Burkhart, Daphne Tan, and Praneeth Namburi. The biomechanics of piano playing: a systematic review of kinematic, kinetic, and electromyographic literature. *Frontiers in Psychology*, Volume 16 - 2025, 2026. 1

[19] Kazuki Kawamura, Fujiki Nakamura, Hayato Nishioka, Momoko Shioki, Shinichi Furuya, and Jun Rekimoto. Interpretable visualization of expertise-dependent motor skills toward supporting piano practice. In *Proceedings of the 2026 Designing Interactive Systems Conference*, page 2868–2885, New York, NY, USA, 2026. Association for Computing Machinery. 2

[20] Yonghyun Kim, Junhyung Park, Joonhyung Bae, Kirak Kim, Taegyun Kwon, Alexander Lerch, and Juhan Nam. Pianovam: A multimodal piano performance dataset. In *Proceedings of the 26th International Society for Music Information Retrieval Conference (ISMIR)*, 2025. 2

[21] Theodoros Kyriakou, Andreas Aristidou, and Panayiotis Charalambous. Multi-Modal Instrument Performance: A musical database. *Comput. Graph. Forum*, 44(2):e70025, 2025. 2

[22] Zechen Li, Shohreh Deldari, Linyao Chen, Hao Xue, and Flora D. Salim. SensorLLM: Aligning large language models with motion sensors for human activity recognition. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 354–379. Association for Computational Linguistics, 2025. 3

[23] Chin-Yew Lin. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain, 2004. Association for Computational Linguistics. 6

[24] Ruofan Liu, Yichen Peng, Takanori Oku, Chen-Chieh Liao, Erwin Wu, Shinichi Furuya, and Hideki Koike. Piamuscle: Improving piano skill acquisition by cost-effectively estimating and visualizing activities of miniature hand muscles. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, New York, NY, USA, 2025. Association for Computing Machinery. 2

[25] RUOFAN LIU, YICHEN PENG, Takanori Oku, Chen-Chieh Liao, Erwin Wu, Shinichi Furuya, and Hideki Koike. From pose to muscle: Multimodal learning for piano hand muscle electromyography. In *Advances in Neural Information Processing Systems*, pages 115558–115587. Curran Associates, Inc., 2025. 1, 2, 3

[26] Seungwhan Moon, Andrea Madotto, Zhaojiang Lin, Alireza Dirafzoon, Aparajita Saraf, Amy Bearman, and Babak Damavandi. Imu2clip: Multimodal contrastive learning for imu motion sensors from egocentric videos and text. *arXiv preprint arXiv:2210.14395*, 2022. 3

[27] Takanori Oku, Masato Hirano, Kengo Matsuzaka, and Shinichi Furuya. Augmented feedback training for overcoming the learning plateau of motor expertise. *bioRxiv*, 2025. 2

[28] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting on Association for Computational Linguistics*, page 311–318, USA, 2002. Association for Computational Linguistics. 6

[29] Paritosh Parmar, Jaiden Reddy, and Brendan Morris. Piano skills assessment. *arXiv preprint arXiv:2101.04884*, 2021. 2

[30] Kumar Ashis Pati, Siddharth Gururani, and Alexander Lerch. Assessment of student music performances using deep neural networks. *Applied Sciences*, 8(4), 2018. 2

[31] Qwen Team. Qwen3.5: Towards native multimodal agents, 2026. 4, 6

[32] Lala Shakti Swarup Ray, Vitor Fortes Rey, Bo Zhou, Paul Lukowicz, and Sungho Suh. Chairpose: Pressure-based chair morphology grounded sitting pose estimation through simulation-assisted training. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*, New York, NY, USA, 2025. Association for Computing Machinery. 2

[33] Tatsuki Seino, Naoki Saito, Takahiro Ogawa, Satoshi Asamizu, and Miki Haseyama. Expert comment generation considering sports skill level using a large multimodal model with video and spatial-temporal motion features. *Sensors*, 25 (2), 2025. 2

[34] Ziwei Shan, Yaoyu He, Chengfeng Zhao, Jiashen Du, Jingyan Zhang, Qixuan Zhang, Jingyi Yu, and Lan Xu. Mojito: Llm-aided motion instructor with jitter-reduced inertial tokens. *arXiv preprint arXiv:2502.16175*, 2025. 3

[35] Xian Shi, Xiong Wang, Zhifang Guo, Yongqi Wang, Pei Zhang, Xinyu Zhang, Zishan Guo, Hongkun Hao, Yu Xi, Baosong Yang, Jin Xu, Jingren Zhou, and Junyang Lin. Qwen3-asr technical report. *arXiv preprint arXiv:2601.21337*, 2026. 12

[36] Davinder Pal Singh, Lala Shakti Swarup Ray, Bo Zhou, Sungho Suh, and Paul Lukowicz. A novel local-global feature fusion framework for body-weight exercise recognition with pressure mapping sensors. In *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 6375–6379, 2024. 2

[37] H.Z. Tan, L.A. Slivovsky, and A. Pentland. A sensing chair using pressure distribution sensors. *IEEE/ASME Transactions on Mechatronics*, 6(3):261–268, 2001. 2

[38] Abhishek Tandon, Anujraaj Goyal, Henry M. Clever, and Zackory Erickson. Bodymap - jointly predicting body mesh and 3d applied pressure map for people in bed. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2480–2489, 2024. 2

[39] Craig Turner, Peter Visentin, Deanna Oye, Scott Rathwell, and Gongbing Shan. An examination of trunk and right-hand coordination in piano performance: A case comparison of three pianists. *Frontiers in Psychology*, Volume 13 - 2022, 2022. 1

[40] Craig Turner, Robin Mailly, Fabien Dal Maso, and Felipe Verdugo. Impact of expressive intentions on upper-body kinematics in two expert pianists. *Frontiers in Psychology*, Volume 15 - 2024, 2025. 1

[41] Aaron van den Oord, Oriol Vinyals, and Koray Kavukcuoglu. Neural discrete representation learning. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, page 6309–6318, Red Hook, NY, USA, 2017. Curran Associates Inc. 3

[42] Felipe Verdugo, Justine Pelletier, Benjamin Michaud, Caroline Traube, and Mickaël Begon. Effects of trunk motion, touch, and articulation on upper-limb velocities and on joint contribution to endpoint velocities during the production of loud piano tones. *Frontiers in Psychology*, Volume 11 - 2020, 2020. 1

[43] Ruocheng Wang, Pei Xu, Haochen Shi, Elizabeth Schumann, and C. Karen Liu. Fürelise: Capturing and physically synthesizing hand motions of piano performance. In *SIGGRAPH Asia 2024*, 2024. 1, 2, 3

[44] Zhiyuan Wang, Zhiqian Xia, Guangyao Zhao, Kaiyue Lu, Haifeng Xia, and Siyu Xia. Generating tennis action instruction based on a large language model. In *2025 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, pages 6616–6625, 2025. 2

[45] Erwin Wu, Takashi Matsumoto, Chen-Chieh Liao, Ruofan Liu, Hidetaka Katsuyama, Yuki Inaba, Noriko Hakamada, Yusuke Yamamoto, Yusuke Ishige, and Hideki Koike. Skitech: An alpine skiing and snowboarding dataset of 3d body pose, sole pressure, and electromyography. In *Proceedings of the 6th International Workshop on Multimedia Content Analysis in Sports*, page 3–8, New York, NY, USA, 2023. Association for Computing Machinery. 2

[46] Erwin Wu, Hayato Nishioka, Shinichi Furuya, and Hideki Koike. Marker-removal networks to collect precise 3d hand data for rgb-based estimation and its application in piano. In *2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 2976–2985, 2023. 1, 2, 3

[47] Qi Wu, Yubo Zhao, Yifan Wang, Xinhang Liu, Yu-Wing Tai, and Chi-Keung Tang. Motion-agent: A conversational framework for human motion generation with llms. In *International Conference on Learning Representations*, pages 48023–48043, 2025. 3

[48] Qingjun Xing, Xuyang Xing, Ping Guo, Zhenhui Tang, and Yanfei Shen. Llm-fms: A fine-grained dataset for functional movement screen action quality assessment. *PLOS ONE*, 20 (3):1–16, 2025. 2

[49] Wei-Hsin Yeh, Yu-An Su, Chih-Ning Chen, Yi-Hsueh Lin, Calvin Ku, Wenhsin Chiu, Min-Chun Hu, and Lun-Wei Ku. CoachMe: Decoding sport elements with a reference-based coaching instruction generation model. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 29126–29151, Vienna, Austria, 2025. Association for Computational Linguistics. 2, 8

[50] Hao Yin, Paritosh Parmar, Daoliang Xu, Yang Zhang, Tianyou Zheng, and Weiwei Fu. A decade of action quality assessment: Largest systematic survey of trends, challenges, and future directions. *International Journal of Computer Vision (IJCV)*, 2026. 2

[51] Huan Zhang, Jinhua Liang, and Simon Dixon. From audio encoders to piano judges: Benchmarking performance understanding for solo piano. In *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*, San Francisco, USA, 2024. 2

[52] Tianyi Zhang*, Varsha Kishore*, Felix Wu*, Kilian Q. Weinberger, and Yoav Artzi. Bertscore: Evaluating text generation with bert. In *International Conference on Learning Representations*, 2020. 6

[53] Xinyu Zhao, Nikita Karagodin, Hamed Hassani, Sinan Hersek, Paul Pu Liang, and Yury Polyanskiy. Continuous first, discrete later: VQ-VAEs without dimensional collapse. *arXiv preprint arXiv:2605.06870*, 2026. 6

[54] Adrian Łańcucki, Jan Chorowski, Guillaume Sanchez, Ricard Marxer, Nanxin Chen, Hans J.G.A. Dolfing, Sameer Khurana, Tanel Alumäe, and Antoine Laurent. Robust training of vector quantized bottleneck models. In *2020 International Joint Conference on Neural Networks (IJCNN)*, pages 1–7, 2020. 6



Figure 6. Score shown for the *Chord LR staccato sharp* coaching task. Participants were instructed to "play with a sharp, resonant sound."

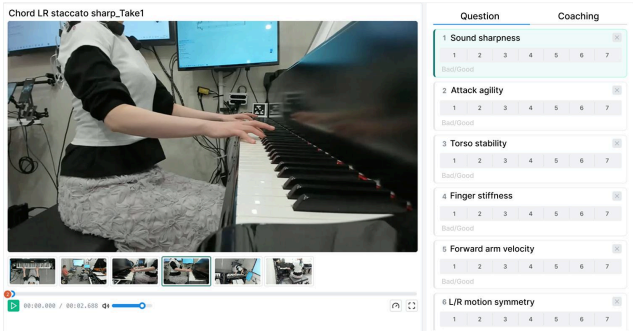


Figure 7. Coaching annotation interface. Annotators could switch among six synchronized views, replay the performance with audio, assign seven criterion-specific ratings on a 1–7 scale, and enter free-form coaching comments.

A. Dataset Details

A.1. Coaching Task and Annotation Interface

Figure 6 shows the score used for the coaching target. The fixed score, tempo, and performance instruction hold the musical condition constant across participants, allowing the experts to focus on differences in execution.

Figure 7 shows the interface used to collect skill ratings and coaching. Annotators independently replayed the synchronized views and audio as needed before submitting all seven ratings and their comments. Comments could be entered as text or speech. Spoken comments were transcribed with Qwen3-ASR-1.7B [35], then reviewed and corrected by the annotator in the same interface before submission. Although the interface also stored a video timestamp for each comment, the target performances were too short for reliable temporal localization; we therefore treat every comment as describing the complete performance.

A.2. Supplementary Video and Sample Dataset

The supplementary video presents concrete examples of the multimodal recordings in PianoCC and representative responses generated by PianoCC-LM. It complements the static figures by showing the recorded modalities over time and illustrating the model's outputs for piano-performance understanding and coaching. We also provide a sample dataset containing a subset of the multimodal recordings and annotations from PianoCC.

Table 5. Writing style and major-label mention rates in expert coaching.

	R1	R2	R3
Comments per performance	1.00	1.00	2.21
Prescriptive expressions	0.17	0.93	1.00
Descriptive expressions	0.98	0.04	0.00
Mention: fingertip	0.87	0.68	0.70
Mention: keying motion	0.78	0.69	0.70
Mention: upper arm / shoulder	0.83	0.43	0.43
Mention: sound quality	0.77	0.08	0.00
Mention: local tension	0.74	0.42	0.30

B. Annotation Analysis Details

B.1. Coaching Style

Table 5 shows that the three experts observed and described each performance from different perspectives. R1 predominantly describes observations, whereas R2 and R3 predominantly prescribe improvements. We therefore retain their coaching comments as independent targets rather than merging them into a single response.

C. QA Dataset Construction Details

C.1. Pressure Description QA

For every pressure interval, we compute total, left–right, and front–back loads and the x and y coordinates of the CoP. Comparing the initial and final segments yields five change levels: *stable*, *slightly increasing*, *increasing*, *slightly decreasing*, and *decreasing*. The CoP and regional loads determine whether pressure is concentrated at the front, back, left, right, or a combination. Templates combine these attributes into factually grounded answers.

C.2. Posture Description QA

We align 30-Hz 3D keypoints with each pressure interval by nearest-neighbor sampling and retain the pelvis, spine chest, and both shoulders. From these joints, we calculate forward/backward and lateral trunk angles, shoulder levelness, and within-interval trunk motion. We use 5° thresholds for trunk lean and stability and a 3° threshold for shoulder levelness. An interval produces four attribute-specific answers and one combined summary; intervals with fewer than half valid frames are excluded.

C.3. Skill Assessment and Coaching QA

Skill ratings are matched to sessions through the recording manifest, and participant identity determines the data split. We average the three ratings, apply half-up rounding, and clip the target to 1–7. For coaching, utterances are concatenated for each rater–performance pair in timestamp order. The *overall* target preserves the complete response, while

individual targets split it into self-contained items without separating clauses that depend on one another. Provenance and rater identity are retained.

D. Experiment Details

D.1. ChairVAE and Language-model Training

ChairVAE treats the 264 pressure cells as feature channels and applies convolution along time. Its vector quantizer contains 1,024 codes of dimension 512. We first optimize the encoder–decoder with Smooth L1 reconstruction loss and then enable vector quantization with

$$\mathcal{L}_{\text{ChairVAE}} = \mathcal{L}_{\text{SmoothL1}} + 0.02\mathcal{L}_{\text{commit}}. \quad (2)$$

The projector is a two-layer MLP. Both language-model stages run for three epochs; Stage 2 uses LoRA rank 16, alpha 32, and dropout 0.05 on Qwen's linear layers. System prompts, questions, and pressure placeholders are masked from the causal-language-modeling loss. Training completes in approximately six hours on one NVIDIA H100 GPU.

D.2. Baseline Settings

The pressure Random Forest uses the mean, standard deviation, minimum, maximum, and three adjacent-frame-difference statistics for every cell. It contains 200 trees with maximum depth 20. Pose variants use the synchronized 27-joint sequence. The Transformer has four encoder layers and is trained separately for task and skill targets.

Nearest Neighbor retrieves the closest training example within the same input modality and reuses its available task, rating, or coaching target. Qwen3.5-9B-FT receives a horizontally flipped 10-fps pressure heatmap video colored with the inferno map and normalized with the training-set 99.5th percentile. Its visual encoder and aligner are frozen, and all-linear LoRA is trained for three epochs. The pose PianoCC-LM uses a pose VQ-VAE with the same temporal compression, projector interface, and two-stage schedule as the pressure model.

D.3. Detailed Classification Results

Table 6 reports task-classification results at both levels of the task taxonomy. The fine-grained setting distinguishes all 40 performance conditions, whereas the coarse setting merges them into the 10 major categories shown in Fig. 2.

Table 7 supplements the seven-level skill accuracy in the main paper with macro-F1 and mean absolute error (MAE). Macro-F1 measures performance across rating classes, while MAE accounts for the ordinal distance between the predicted and target ratings.

Table 6. Task-classification results for the 40 performance conditions and 10 major task categories.

Method	Input	40 Classes		10 Classes	
		Acc.	F1	Acc.	F1
Random Forest	Pose	**0.646**	0.620	**0.923**	0.881
Transformer	Pose	0.614	0.589	0.912	**0.889**
Nearest Neighbor	Pose	0.382	0.329	0.516	0.528
Qwen3.5-9B	Pose	0.022	0.022	0.112	0.132
Qwen3.5-9B-FT	Pose	0.412	0.342	0.598	0.622
Random Forest	Press.	0.404	0.356	0.768	0.742
Transformer	Press.	0.337	0.283	0.751	0.692
Nearest Neighbor	Press.	0.232	0.232	0.486	0.477
Qwen3.5-9B	Press.	0.026	0.003	0.122	0.072
Qwen3.5-9B-FT	Press.	0.114	0.080	0.362	0.340
PianoCC-LM	Pose	0.632	0.619	0.901	0.845
PianoCC-LM	Press.	0.390	0.354	0.752	0.771
PianoCC-LM	Press.+Pose	0.642	**0.622**	0.913	0.880

Table 7. Seven-level skill-assessment results. Accuracy values are reproduced from the main paper; F1 and MAE will be added after aggregation.

Method	Input	Acc.	F1	MAE
Random Forest	Pose	0.333	0.321	0.961
Transformer	Pose	0.341	0.329	0.974
Nearest Neighbor	Pose	0.262	0.231	1.318
Qwen3.5-9B	Pose	0.301	0.285	1.075
Qwen3.5-9B-FT	Pose	0.294	0.289	1.012
Random Forest	Press.	0.318	0.312	0.894
Transformer	Press.	0.363	0.351	0.874
Nearest Neighbor	Press.	0.286	0.250	1.253
Qwen3.5-9B	Press.	0.276	0.217	1.323
Qwen3.5-9B-FT	Press.	0.285	0.240	1.271
PianoCC-LM	Pose	0.375	0.353	0.714
PianoCC-LM	Press.	**0.388**	0.362	**0.692**
PianoCC-LM	Press.+Pose	0.382	**0.380**	0.697